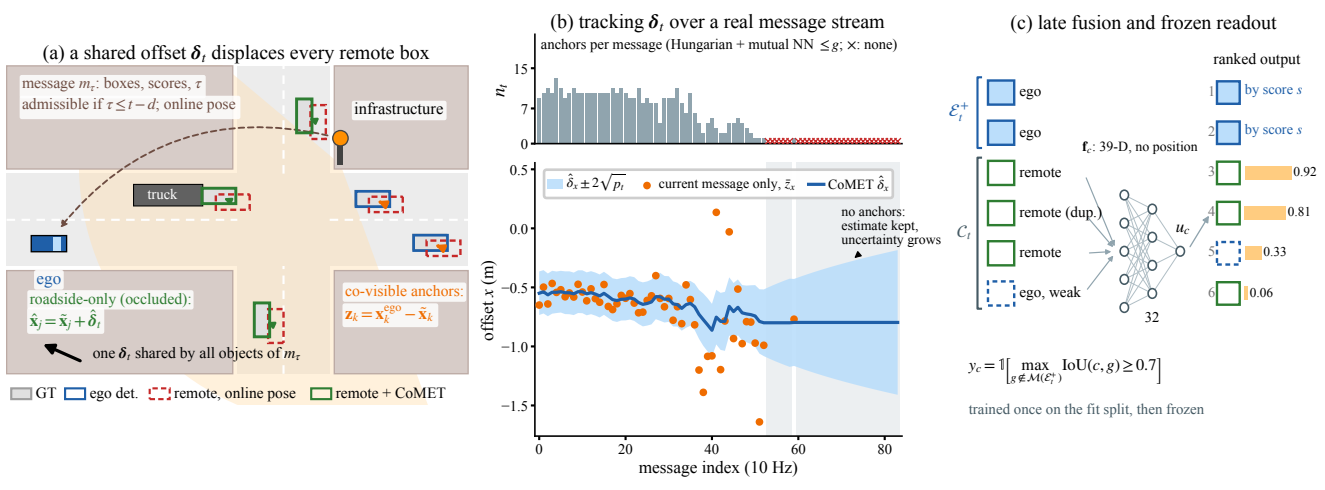

(a) a shared offset $\boldsymbol{\delta}_t$ displaces every remote box
message m_τ: boxes, scores, τ
admissible if $\tau \le t-d$; online pose
infrastructure
truck
ego
roadside-only (occluded):
$\hat{\mathbf{x}}_j = \tilde{\mathbf{x}}_j + \hat{\boldsymbol{\delta}}_t$
co-visible anchors:
$\mathbf{z}_k = \mathbf{x}_k^{\text{ego}} - \tilde{\mathbf{x}}_k$
one $\boldsymbol{\delta}_t$ shared by all objects of m_τ
GT
ego det.
remote, online pose
remote + CoMET
(b) tracking $\boldsymbol{\delta}_t$ over a real message stream
anchors per message (Hungarian + mutual NN $\le g$; ×: none)
n_t
15
7
0
$\hat{\delta}_t \pm 2\sqrt{p_t}$
current message only, $\tilde{z}_t$
CoMET $\hat{\delta}_t$
no anchors:
estimate kept,
uncertainty grows
offset x (m)
0.5
0.0
−0.5
−1.0
−1.5
0
20
40
60
80
message index (10 Hz)
(c) late fusion and frozen readout
ranked output
$\mathcal{E}_t^+$
ego
ego
$\mathcal{C}_t$
remote
remote (dup.)
remote
ego, weak
$\mathbf{f}_c$: 39-D, no position
u_c
32
1
by score s
2
by score s
3
0.92
4
0.81
5
0.33
6
0.06
$y_c = \mathbb{1}\left[\max_{g \notin \mathcal{M}(\mathcal{E}_t^+)} \text{IoU}(c,g) \ge 0.7\right]$
trained once on the fit split, then frozen